

FROM HYDROPOWER TO COMPUTE

NEPAL'S NARROW WINDOW TO BECOME AN
AI INFRASTRUCTURE EXPORT HUB

WORKING DRAFT

Authors

Shreeansh Agrawal
Akanksha Baidya

Strategic Advisor

Bibek Raj Kandel

info@cantor-dust.com



Draft Notice

Working Draft Notice

This paper is a working draft prepared for discussion, review, and refinement. The analysis, assumptions, figures, recommendations, and conclusions contained herein are subject to revision as additional data, stakeholder feedback, regulatory clarification, and technical validation become available.

Readers should not treat this document as a final policy position, investment memorandum, legal opinion, or technical feasibility report. It is intended to support informed debate on Nepal's potential role in AI infrastructure and compute exports.

Comments, corrections, and technical feedback are welcome and may be incorporated into future versions of the paper.

Disclaimer

This document has been prepared for informational and discussion purposes only. The views expressed are those of the authors and do not necessarily represent the views of any government agency, regulator, investor, technology provider, or other institution mentioned in the paper.

The analysis relies on publicly available information, author estimates, and stated assumptions. While reasonable efforts have been made to ensure accuracy, Cantordust and the authors make no representation or warranty, express or implied, as to the completeness, accuracy, or reliability of the information contained herein. Readers should independently verify any data, assumptions, financial projections, regulatory interpretations, or policy recommendations before relying on them.

Nothing in this document should be construed as legal, financial, tax, investment, engineering, or regulatory advice.



About Cantordust and Himalayan Sanctuary Initiative

This paper was developed through a collaboration between Cantordust and the Himalayan Sanctuary Initiative (HSI).

Cantordust is an AI and analytics firm focused on building bespoke solutions for complex industries, with an emphasis on practical, mission-aligned AI implementation rather than generic automation. The firm works across sectors including energy, climate, healthcare, manufacturing, and international development, combining data science, machine learning, economic analysis, and sector-specific research.

The Himalayan Sanctuary Initiative is a think tank and research organization dedicated to advancing sustainable development, environmental governance, and climate-resilient infrastructure in the Himalayan region. HSI's work spans energy policy, watershed management, institutional governance, and digital infrastructure in frontier economies.

Copyright and Use

© 2026 Cantordust. All rights reserved.

No part of this document may be reproduced, distributed, modified, or transmitted in any form without prior written permission from Cantordust, except for brief quotations used for review, commentary, research, or policy discussion with appropriate attribution.

For permissions, comments, or inquiries, please contact:

info@cantor-dust.com



Table of Contents

Draft Notice	i
Executive Summary.....	iv
Introduction	1
Data Centers 101	4
Nepal’s Data Outlook	6
Nepal’s Regulatory Landscape	11
Global Outlook	17
The Case for Nepal as a Compute Export Hub	19
What to Build: Anticipating the Next Wave of Compute Demand	21
Financial Modeling	23
Conclusion	31



Executive Summary

While the richest nations on earth face a crippling energy crunch to power the artificial intelligence boom, Nepal is uniquely positioned to fuel the future of global technology. The math is simple. AI needs power, and Nepal has it. Tech giants are spending hundreds of billions of dollars to build infrastructure, but they have hit a brick wall. A landmark [International Energy Agency \(IEA\) report](#) reveals that global data center electricity demand jumped by 17% in 2025 alone, while power consumed by AI facilities soared by 50%. This global energy crisis is Nepal's historic opportunity.

Nepal's new, Balendra Shah led, post-Gen Z government has been vocal about this too. Recognizing the shift toward a digital future, the Balen administration has explicitly built this aspiration into its core agenda, embedding it directly within its 100-point program.

But this aspiration has a strictly limited window, and there is still so much to be done. Before the global tech movements look elsewhere, we need to move at lightning speed, as many competing candidates from Kenya to Malaysia lock down these contracts, leaving Nepal's energy wealth stranded. To turn this political will into a real-world tech boom, we must immediately leverage our strengths:

1. a hydropower surplus that maps onto the clean, firm-power profile that hyperscale buyers are now paying premiums for
2. ambient temperatures and water access that materially reduce cooling overhead
3. an industrial electricity tariff (roughly \$0.05/kWh) that runs up to 50% below comparable US rates.

While significant, these advantages are not sufficient on their own to warrant investment.

A representative pilot facility — a 512-GPU NVIDIA H100 cluster drawing approximately 0.6 MW, with a total CAPEX of \$41.9 million — illustrates the gap between potential and viability. Under the current policy regime, the project returns a negative NPV. Under a reformed regime (zero customs, concessional financing), the same project returns a positive NPV. The single decisive variable is the import duty. Across the 72-scenario sensitivity matrix, every configuration that retains the full 20% customs rate produces a negative NPV, regardless of corporate tax treatment or financing cost. Conversely, every configuration with zero customs duty is positive. Customs reform is therefore a binding constraint.



Beyond that, three things must happen for Nepal to participate in this market:

1. a sector-specific customs exemption for compute hardware imported for export-oriented compute services
2. explicit regulatory clarity on cross-border data flows, governing law, and the operational status of the Data Protection Authority
3. PPA framework that addresses the seasonal profile of run-of-river generation, which a 24/7 training facility cannot otherwise absorb.

None of these requires new legislation. All of them require focused political prioritization within the next twelve to eighteen months, before competing jurisdictions (Bhutan, the Gulf, Southeast Asia) and rapidly emerging Sub-Saharan African economies close their own positioning gaps.

The remainder of this paper develops the case in three parts: a primer on the data center landscape and Nepal's current digital footprint; a global-context discussion of why training workloads in particular are a perfect match for Nepal's clean-energy surplus; and a financial model that quantifies exactly how the proposed policy reforms will unlock the investors return.

Introduction

The world's artificial intelligence labs are running short on power. Microsoft recently restarted a nuclear reactor to secure electricity for its data centers. Google is building enhanced geothermal facilities from scratch. Amazon is negotiating multi-year contracts for every available megawatt of clean power in North America and Europe. Meanwhile, Nepal, a country that generates more hydroelectric power than it consumes is positioned to capture a fraction of this demand but risks losing the opportunity entirely to faster-moving competitors if the government does not act within the next 12 to 18 months.

The clean energy debate in Nepal has recently revolved around a single question: how to absorb the surplus generated by accelerating hydropower buildout without exporting it at depressed wholesale rates to a single buyer. Industrial electrification, electric mobility, and cross-border power trade have each been advanced as partial answers. Data centers have now joined that list, and they bring a fundamentally different proposition: rather than exporting electricity, Nepal would export the computation that electricity produces, capturing margin that currently flows to data center operators in higher-cost jurisdictions.

The momentum behind this idea is broad. The Government of Nepal has issued the [Data Centre and Cloud Services Directive 2081](#) to establish a licensing and Tier-classification framework. Senior policymakers have [publicly positioned Nepal as a candidate destination](#) on the basis of clean energy and natural cooling. The Rastriya Swatantra Party, which won a near-supermajority in the March 2026 elections, has made compute-export a named pillar of its economic platform. Domestic operators including [WorldLink](#) have announced commercial buildouts, and at least two foreign-anchored projects (Yotta-BLC, Huawei-Nepal Telecom) are in development.

This convergence, while exciting, has also produced a degree of conceptual flattening and has created a dangerous policy blind spot. In Nepal today, the term "Data centers" is thrown around indiscriminately to describe a small bank server cabinet, a government cloud facility (Huawei-built), and the kind of multi-megawatt AI training facility that Microsoft and Google are now building behind dedicated nuclear PPAs.

This paper argues for a narrower and more defensible proposition: Nepal's structural advantages are best suited for latency-tolerant AI training workloads. AI training jobs can run continuously for weeks or months, consume large volumes of baseload electricity, and are less dependent on proximity to end-users than real-time inference or consumer cloud services.



However, potential is not the same as viability. Nepal's existing digital infrastructure is still modest. Most domestic compute demand is limited to banking systems, government e-services, ISP infrastructure, CDN caching, and enterprise server rooms. A rough top-down estimate suggests that total compute-related energy demand linked to Nepal is unlikely to exceed 20–25 MW today, and a significant share of heavier dynamic compute still takes place in India, Singapore, or other offshore cloud regions. Incremental growth in domestic digital consumption will likely be absorbed by existing or already planned facilities. The larger opportunity is therefore not local demand substitution, but export-oriented AI compute.

The global context makes this opportunity strategically relevant. AI training and inference are expected to account for a growing share of data center electricity demand, even though forecasts differ widely because of uncertainty around model efficiency and adoption rates. Hyperscalers and AI labs are already treating clean, firm power as a strategic input. Major technology companies are signing long-term power purchase agreements tied to nuclear, geothermal, and other carbon-free electricity sources. This signals a structural shift: compute buyers are willing to pay for reliable clean energy when it secures long-duration AI infrastructure.

Nepal's hydropower base gives it a credible entry point into this market, but only if the country solves several institutional constraints. The Data Centre and Cloud Services Directive 2081 provides a starting framework, but it was primarily designed for domestic ICT regulation rather than export-grade AI infrastructure. It does not yet resolve questions around cross-border data flows, international customer rights, governing law, breach notification, or globally recognized compliance standards. But customs reform alone is insufficient. Three complementary institutional reforms are required, all implementable within 12–18 months through executive guidance and inter-agency coordination:

1. **Data Protection Authority operationalization.**
2. **Firm-power Power Purchase Agreement framework with NEA.**
3. **Cross-border data flows and governing law clarity.**

These three reforms, combined with customs exemption, create the conditions for viability. Without all four, Nepal's cost advantage remains theoretical. With all four, the pilot facility becomes commercially viable.



The financial model developed in this paper makes that last point explicit. We model a representative 512-GPU NVIDIA H100 cluster, equivalent to roughly 0.6 MW of continuous load, across 72 scenarios varying customs duty, corporate tax, cost of debt, and electricity tariff. Under the current policy regime, the base case produces a negative NPV and an IRR below the estimated WACC. The result is not primarily driven by electricity cost, concessional financing, or corporate tax treatment. It is driven by import duty on GPU hardware. Every scenario retaining the current high effective import duty produces a negative NPV. Every scenario with a customs waiver produces a positive NPV. The model implies that a sector-specific customs exemption for GPU hardware imported for export-oriented compute services is the non-negotiable first condition for viability.

The timing matters. The H100 generation that anchors the financial model has a finite commercial life before successor architectures such as Blackwell and Rubin reshape GPU economics. Meanwhile, Bhutan, Gulf countries, and several Southeast Asian jurisdictions are positioning aggressively for the same compute-export opportunity with deeper capital pools and faster regulatory cycles.

The remainder of this paper develops the argument in sequence. It begins by distinguishing among data center categories and clarifying why AI training is the relevant opportunity for Nepal. It then maps Nepal's current digital infrastructure, bandwidth architecture, and compute-related energy demand. The paper next examines the regulatory landscape and identifies the institutional gaps that must be resolved for export-oriented compute. It then situates Nepal within the global AI compute buildout, explains the structural case for Nepal as a clean compute hub, and presents the financial model showing which policy levers determine viability. The paper concludes with a practical reform package: customs exemption for GPU hardware, an industrial-rate firm-power PPA pathway, operationalization of data protection oversight, and cross-border data governance clarity.

Data Centers 101

To center the discussion on potential interventions in the Nepali landscape, we must first converge on what data centers are.

Under the most general definition, data centers are specialized facilities designed to house computing infrastructure. At their core, they serve three essential functions:

1. ensuring uninterrupted operation of servers
2. maintaining optimal thermal conditions through reliable cooling systems
3. enabling high-speed connectivity to telecommunications networks.

In Nepal's current infrastructure reality, most organizations operate servers within office premises or rely on small-scale colocation facilities (defined below). These arrangements are inherently difficult to scale and often fall short of international reliability standards due to three critical gaps:

1. absence of dedicated and stable power supply
2. limited redundancy in cooling systems in the event of HVAC failure
3. insufficient operational oversight and monitoring.

Data centers can be broadly categorized into three types:

- 1) **Enterprise (private) data centers:** These are owned and operated by individual organizations to support their internal IT infrastructure. This remains the most common model in Nepal today.
- 2) **Colocation (colo) data centers:** These facilities lease rack space to multiple customers while providing shared services such as backup power, cooling, physical security, and network connectivity. They offer a more efficient and reliable alternative to in-house server setups.
- 3) **Cloud or hyperscale data centers:** Operated by major cloud service providers such as AWS, Azure and GCP, these facilities operate at significantly larger scales and serve millions of users globally. In the Nepali context, most businesses accessing cloud services are connected to regional hubs such as Mumbai or Southeast Asia, rather than domestic infrastructure.

Data centers are also commonly classified according to standardized "Tier" ratings, which reflect their level of redundancy, fault tolerance, and expected uptime.¹ Any

¹ For detailed breakdown, refer to the [Uptime Institute's Tier Classification](#) Standards.

project aimed at exporting services should target at least Tier III to be commercially viable, as this is the de facto industry minimum for hyperscale and enterprise customers.

Two compute markets exist with vastly different infrastructure demands and economics:

1. **General compute** (web hosting, databases, applications, services) requires standard cooling and latency tolerances. This is what dominates current Nepal discussions because it is more familiar and requires lower upfront capital.
2. **AI-focused compute** (model training, inference, LLM services) requires sophisticated liquid cooling, massive power density, and zero-latency local connectivity between GPU clusters. These facilities are far more capital-intensive than general compute facilities but serve much larger addressable markets globally. A single regional AI hub in South Asia could capture significant economic value.

Table 1: Breakdown of Compute Uses. Source: Authors

		Description	Hardware	Energy Use	Duration
General Compute >>		Everything a conventional data center does: web servers, database querying, email, file storage, transaction processing, ERP, etc.	CPU Servers	Low: 300-600 W per Server 5-15 kW / rack	Continuous. Latency requirements depend on use case.
AI/ML >>	Inference	Using a trained model to generate output. This is what happens when ChatGPT is responding to your prompt.	GPU clusters	Medium: 150-700 per GPU 10-40 kW / rack	Continuous. Latency must be very low.
	Training	Building a model from scratch: feeding billions of examples through a model until it learns patterns. Requires large-scale deployment planning	GPU clusters	Extremely High: H100s draw 700W each 50-100+ kW / rack	Highly concentrated bursts. Takes days to run.

A **critical planning implication is the following:** current Nepal discussions center on general computing because it is easier to conceptualize. However, if Nepal targets a Tier III facility, the strategic question becomes: should the facility be designed from inception to support *both* general compute *and* AI-focused compute? If Nepal positions itself to capture AI infrastructure workloads at Tier III scale, it could become a South Asian hub for this much larger opportunity, but only if the cooling, power, and latency architecture are designed for AI from day one. Building for general compute alone risks creating infrastructure that cannot economically upgrade to AI-grade specifications later.

Nepal's Data Outlook

Nepal does not have direct access to submarine cable systems and therefore relies on cross-border terrestrial links for international internet bandwidth. The vast majority of this bandwidth is routed through India, with key entry points at Biratnagar, Birgunj and Bhairahawa. A smaller share is routed through China via the Rasuwagadhi-Kerung land border, established in 2018, with onward transit through Chinese networks.



Figure 1: Major Fiber Backbone and Exits. Source: Worldlink

Once in Nepal, international bandwidth is received by Network Service Providers (NSPs), companies licensed by the NTA to purchase international gateway capacity and wholesale it to internet service providers. Traffic is distributed domestically through optical fiber infrastructure, including the Nepal Electricity Authority's Optical Ground Wire (OPGW) network — which integrates fiber-optic cables with high-voltage transmission lines along NEA corridors — and the Nepal Telecom-operated Mid-Hill optical fiber backbone, which connects district headquarters across the country.

Kathmandu hosts Nepal's Internet Exchange Point (NPIX), the central facility where major ISPs interconnect and domestic traffic is redistributed. To reduce dependence on international links for high-volume consumer traffic, operators such as WorldLink deploy Content Delivery Networks (CDNs) that store frequently accessed content locally, including from Google, Facebook, and Youtube, on servers within Nepal. When users

request supported content, it is served locally without traversing international links. Lastly, Internet Service Providers (ISPs) like Subisu, Worldlink, NTC connect to NPIX and then use the same Optical Ground Wire (OPGW) network and their own wire network to deliver internet to the user.

Importantly, none of the domestic/distributive architecture is optimized for high-intensity GPU workloads, which would require dedicated bandwidth. This architecture is outlined in Figure 2:

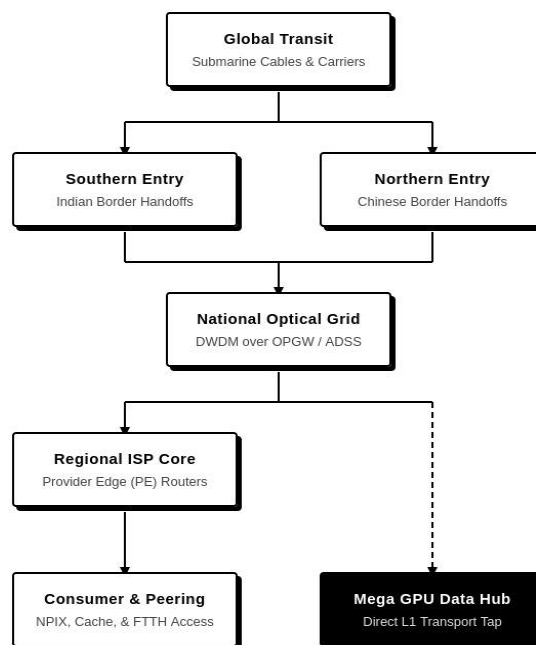


Figure 2: Flowchart of internet architecture. Source: Authors

Both digital demand and supply remain underdeveloped, even for relatively modest CPU-based workloads. Existing capacity is concentrated in a limited number of colocation and enterprise facilities, alongside a [government-operated data center](#) in Kathmandu. We list some of these facilities in Table 2 based on publicly available information and inquiries:

Table 2: List of major players with specifications. Colors indicate viability for GPU services. Source: Authors

Provider / facility	Location	AI Inference Workloads	AI Training Workloads	Tier	Racks	Power (MW)
Data World	Chandragiri; Various Others	Yes; capable of high-power rack densities and advanced air cooling.	Maybe; GPU density is not public.	III	520	3.5
Ncell	Nakkhu	Maybe; can do light inference	Likely Not; No info about GPU clusters or high-density AI racks.	III	250	5
DataHub	Kathmandu, Butwal	No	No; but working to expand into GPU clusters	II	110	Unknown; likely MW scale
Cloud Himalaya	Kathmandu	Yes	Yes; Has VGPU services	III	60	0.5
Dataspace Putalisadak (PTS)	Kathmandu	Maybe; can do light inference	Likely Not; positioned as a telco hotel, not a high-density AI Site	III	14	Unknown; likely kW scale
Silver Lining	Kathmandu		Likely Not; Public rack offer shows 42U full rack with up to 3 KVA power, which is enough for traditional servers, but not a sign of dense GPU training infrastructure		60	0.5

Nepal's existing data center capacity is built almost exclusively for general-compute workloads (banking core systems, government e-services, ISP infrastructure, CDN caching) and is not provisioned for AI training or inference. Additionally, these centers likely are not responsible for the heavier compute that goes behind commercial use cases, since those players deploy private servers. The energy use from compute in Nepal can be broadly categorized into three buckets:

- 1) Content cached locally in Nepal:** Major CDNs like Google (YouTube, Search), Meta (Facebook, Instagram, WhatsApp), Akamai and Cloudflare deploy cache nodes directly inside ISP networks in countries like Nepal. WorldLink explicitly states on its own website that it hosts Google, Facebook, and Akamai cache servers locally. This means a significant fraction of consumer traffic is served from servers physically in Nepal or at the Kathmandu internet exchange (NPIX).
- 2) Dynamic compute in Indian/Singaporean facilities:** A lot of activity that requires real computation (API calls, database queries, fintech transaction processing, cloud services, SaaS applications, AI inference) runs in data centers in Mumbai, Pune, Singapore, or further afield. For some of Nepal's workload mix, this is a prominent category by compute intensity.
- 3) Domestic compute in Nepal's named data centers:** The named DC capacity in Nepal hosts: NRB clearinghouse and connectIPS, major banks' core banking systems, government e-services, some ISP infrastructure, and enterprise server rooms. This is critical compute but relatively small.

Total energy consumption attributable to digital infrastructure in Nepal is difficult to measure directly. A top-down estimate can be derived using internet traffic:

- Reported internet traffic: ~1.2 exabytes annually ([Ken Research, 2023](#))
- Approximation method:

$$\text{Total Traffic} = (\text{Mobile} + \text{Broadband Subscribers} \times \text{Avg GB per user/month} \times 12)$$

Below, we estimate this total traffic and its subsequent energy impact:

Table 3: Estimation of Total Compute Load. Source: Authors

Nepal Compute Demand			
Input / Calculation Step	Central estimate	Unit	Source & notes
A. TRAFFIC INPUTS (physically observed)			
A1 Mobile broadband subscribers (FY 2081/82)	26.5	million	Source: NTA Annual Report 2081/82 (FY 2024/25), Chapter 5 data table, p.56: Mobile Broadband = 26,500,000 (central; note this appears rounded).
A2 Fixed broadband physical connections (FY 2081/82)	3.25	million	Source: NTA Annual Report 2081/82, p.56: Fixed Broadband = 14,196,705 subscriber-equivalents. NTA footnote (MIS reports): '1 Connection equal to 4.37 subscribers for fixed broadband. Physical connections = 14,196,705 ÷ 4.37 = 3,249,817. IMPORTANT: The 14.2M figure represents ~218% household penetration if taken at face value: clearly wrong. 3.25M physical connections = ~50% of Nepal's ~6.5M households, concentrated in Kathmandu Valley, terai cities, and district hqs. Plausible given WorldLink alone claims 1M+ FTTH connections.
A3 Avg mobile data per subscriber/month	4.5	GB/sub/month	Ncell 20-year anniversary press statement (Nepal Telecom, Sep 2025): 'per capita data consumption in Nepal is 4.5 GB per month'. See: https://www.nepaltelecom.com/2025/09/ncell-completes-20-years-leading-charge-for-connectivity.html
A4 Avg fixed broadband consumption per physical connection/month	50	GB/conn/month	No official data available; India's Mobile broadband is 24 GB/sub/month (https://www.pib.gov.in/PressReleasePage.aspx?PRID=2206477&lang=1&reg=3) while fixed broadband averages at ~400 GB /conn/month (https://www.voicendata.com/broadband/fixed-broadband-the-unmatched-backbone-of-a-digital-economy-7368234). So fixed broadband is ~17x mobile. Applying that ratio to Nepal, this would benchmark usage at 4.5 x 17 ~ 75 GB / month. However, Indian usage density is likely higher than Nepal's, and our figures have been adjusted downward accordingly.
B. DERIVED TRAFFIC (calculated)			
B1 Mobile annual traffic	1.43	EB/year	Derived from A1 x A3 x 12 months x unit conversion
B2 Fixed broadband annual traffic	1.95	EB/year	Derived from A2 x A4 x 12 months x unit conversion
B3 Total annual traffic	3.38	EB/year	Cross-check: Ken Research Nepal Telecom Market (2024) reports ~1.2 EB for 2023. Central estimate should be in this order of magnitude. See: https://www.kenresearch.com/nepal-telecom-market
C. ENERGY INTENSITY COEFFICIENT (from literature)			
C1 Server-side energy intensity (electricity per GB of traffic served)	0.06	kWh/GB	Aslan, J., Mayers, K., Koomey, J.G. and France, C. (2018). Electricity Intensity of Internet Data Transmission: Untangling the Estimates. <i>Journal of Industrial Ecology</i> , 22: 785-798. https://doi.org/10.1111/jiec.12630
D. TOTAL COMPUTE ENERGY SERVING NEPAL (wherever hosted)			
D1 Annual server energy	203	GWh/year	B3 (EB) x C1 (kWh/GB) x 10 ⁹ (GB/EB) = kWh; ÷ 10 ³ = GWh. This is total energy consumed globally by servers that serve Nepal's traffic.
D2 Average sustained total compute load	23	MW	D1 (GWh) ÷ 8,760 hrs = GW; x 1,000 = MW. This is the total server load serving Nepal, spread across data centers in India, Singapore, US etc. This is NOT Nepal's domestic DC load.
E. KEY CAVEATS AND LIMITATIONS			
Bandwidth ≠ compute	Mytton et al. (2024) <i>J. Industrial Ecology</i> : energy is NOT proportional to data volume. The kWh/GB coefficient is an average that varies with utilisation, workload type, and facility age. Using a fixed coefficient introduces systematic error — potentially ±50% on energy estimates.		
Traffic figure quality	Nepal does not publish official aggregate internet traffic statistics. The 4.5 GB/month figure is a single operator (Ncell) press statement. NTA MIS reports give subscriber counts, not traffic volumes.		
No AI adjustment	206 Nepal workloads are ~0% AI inference. If AI compute onshores, the kWh/GB coefficient could rise significantly (GPU servers consume 5–10x more per workload than general-purpose servers). This model does not capture AI-driven demand step-change.		

Based on this rough estimate and benchmarking against global energy-per-byte estimates, total compute-related energy demand in Nepal is unlikely to exceed **20 to 25 MW**. Importantly, this figure includes both domestic and offshore compute. Given Nepal's limited data center capacity, a substantial share of compute demand is met outside the country.

Looking ahead, compute-related energy demand in Nepal is likely to grow along two primary dimensions:

Local Digital Consumption Growth:

Rising incomes, increased internet penetration, and expansion of digital services will drive higher demand across both residential and commercial segments. However, given



the small footprint today, this growth is unlikely to make a systemic impact on the total energy profile of the country and will likely be met by existing data centers upgrading their facilities. The theoretical opportunity is already becoming concrete. Nepal is attracting major global tech infrastructure investment: [Google is partnering with Bichuten Data Vault](#) to establish world-class data centers with cloud computing, secure storage, and ML capabilities; this in addition to earlier proposal that include a [Huawei-built data center for Nepal Telecom](#), as well as a much-discussed [4 MW Yotta-BLC data center](#) in Kathmandu. These are not marginal infrastructure projects they signal that global platforms are now willing to invest in Nepal as a regional hub, not merely as a market for external services.

AI and Advanced Compute:

The rapid increase in computational requirements for AI training and inference presents a potential opportunity for Nepal to leverage its clean energy surplus. This is particularly relevant as traditional IT services face disruption from automation and generative AI, and because we have the bandwidth flexibility to better utilize existing infrastructure. We estimate that Nepal currently uses about 1.6 Tbps of bandwidth (assuming older 100 Gbps per wavelength and standard DWDM), while the 96-core fiber backbone it is building can theoretically carry orders of magnitude more. Demand spikes during major national events such as Dashain, Tihar, and the New Year period compress this headroom temporarily, but outside these windows the surplus is substantial. This means that connecting a large compute facility to international customers does not require building new international links from scratch, but rather utilizing capacity that already exists and is already paid for.

Nepal's Regulatory Landscape

Nepal's government has made an explicit strategic commitment to expanding digital infrastructure, including data centers and cloud services, as part of its broader digital transformation agenda. [The Digital Nepal Framework 2.0](#) (2025), which supports the country's broader development planning objectives aligned with the 16th National Development Plan, identifies data centers as critical digital infrastructure and explicitly prioritizes "investing in next-generation infrastructure such as 5G, broadband, and data centres" to "create an environment conducive for innovation and entrepreneurship." Operationally, the government approved the Data Centre and Cloud Services Directive 2081 in January 2025, establishing the licensing framework, Tier classification system, security requirements, and provider-customer engagement standards. This policy foundation is concrete and aligned with Nepal's broader digital agenda. This alignment is important because it situates digital infrastructure development within Nepal's broader national planning framework rather than treating it as a standalone ICT initiative.

However, the existence of a policy framework does not ensure the operational clarity required for private investment. The primary time-sensitive factor is the technological and competitive context. The H100 generation of GPUs anchors the financial model in this analysis for 3-5 years (2026-2031). This generation of GPUs is already considered old. As next-generation GPU architectures emerge and cost-performance dynamics evolve, the underlying economics of AI infrastructure investment will also shift. During this window, several jurisdictions, including countries in the Gulf and parts of Southeast Asia, are positioning more aggressively with clearer regulatory frameworks. Nepal must establish operational clarity on four institutional domains before it can credibly compete for investment.

These gaps are primarily operational instead of legislative: the actual procedures, timelines, institutional responsibilities, and workforce capacity that foreign investors require in contracts and that will determine whether the facility operates effectively after launch. Many of these gaps could potentially be addressed through executive guidance, supplementary directives, and inter-agency coordination within 12-18 months if prioritized.



Institutional Gap 1: Data Protection Authority Not Operationalized

The Directive 2081 establishes operational requirements for data center providers—security standards, Tier certification, breach notification obligations, annual audits, and provider-customer agreements. However, it does not specify the institution responsible for enforcement or the procedures for enforcement.

Section 9 assigns oversight duties to "the Department" (Ministry of Communication and Information Technology) and "the Center" (National Cyber Security Center), but neither is designated as the data protection authority with investigative mandate. The Privacy Act 2075, enacted in 2018, establishes privacy rights for individuals (sections 3-27) but creates no named institution to enforce them. The Data Act 2079, enacted in 2023, establishes a National Statistics Office and Statistics Council for data collection and publication, but not data protection enforcement. Section 6(5) of Directive 2081 requires data center operators to "inform the regulatory body and the center" of unauthorized access or breaches. The regulation does not specify which institution receives the report, the timeline for response, the procedures for investigation, or the mechanisms for customer notification and remedies.

This ambiguity is consequential for private operators. Foreign customers managing GDPR-regulated, CCPA-regulated, or equivalent data may face significant legal and compliance uncertainty when executing data processing agreements without contractual clarity on breach response. They require:

1. a named, empowered authority with explicit investigative mandate
2. published procedures and response timelines (e.g., acknowledgment within 48 hours, initial investigation within 30 days);
3. explicit customer notification rights
4. mechanisms for customer remedies.

The World Bank's paper [Public Institutions in the Age of AI: Emerging Practice](#) emphasizes "Establishing Transparent Oversight" as foundational practice. The framework states: "In the absence of visible and credible oversight, technology in government runs the risks of being perceived as arbitrary, unaccountable, and potentially unjust." For export-oriented infrastructure, this principle applies directly. Without visible, credible oversight authority, foreign customers will not place sensitive workloads in Nepal.



International precedent is instructive. [Singapore operationalized its Personal Data Protection Commission \(PDPC\)](#) within months of enacting its protection law. South Korea's DPA took 18 months from legislation to operation. Vietnam's took two years. Nepal can establish operational clarity (designation of authority, published procedures, timeline commitments) within 6-9 months through a circular or supplementary directive.

Without this clarity, Nepal cannot credibly market itself to international AI labs. This is a contract pre-condition, not a "nice-to-have."

Institutional Gap 2: Electricity Contracting Framework Does Not Address 24/7 Flat-Load Compute

Nepal's Electricity Tariff Collection Regulations 2078 (2021) establish industrial pricing through time-of-day (TOD) tiers: peak (6 hours), normal (12 hours), and off-peak (6 hours). This structure is optimized for traditional industrial users whose power draw varies by shift and production schedule. A 24/7 GPU compute facility draws constant, flat power (approximately 0.6 MW continuously, 365 days per year). The TOD structure offers no advantage to a continuous-load operator and creates contractual complications.

More fundamentally, Nepal's generation profile creates a structural constraint. Nepal's hydropower is predominantly run-of-river generation, meaning output follows rainfall and river discharge rather than demand or operator control. During monsoon season (June-September), generation reaches 60-70% of annual flow, far exceeding domestic demand. During dry season (December-March), generation drops to 20-30% of annual flow. A 0.6 MW facility requesting firm power cannot absorb this seasonal variability without either negotiating a seasonal Power Purchase Agreement with Nepal Electricity Authority (NEA) in which NEA commits firm power during dry season at a negotiated premium importing dry-season power (compromising the "clean energy" value proposition).

The standard NEA PPA template is designed for smaller loads with inherent flexibility; it does not address continuous firm-power requirements. This is a first-order operational constraint—more binding than tariff rates—because without guaranteed 24/7 power availability, the facility cannot meet customer SLAs. No AI lab will commit training workloads to a facility that risks capacity loss.

Resolving this requires NEA to publish supplementary guidance for high-availability industrial contracts specifying seasonal availability bands with corresponding pricing;



standby capacity guarantees; and pricing differentiation between firm and interruptible power. This coordination between NEA, the Ministry of Energy, and the Ministry of Communication and Information Technology is achievable within 4-6 months if designated as priority.

Institutional Gap 3: GPU Import Duty Treatment Undefined

The financial model underlying this paper demonstrates that import duty is the decisive financial variable. Every scenario retaining the current 20% effective import duty produces negative NPV; every scenario with zero duty yields positive NPV.

However, Nepal's current duty on data-processing machines (HS codes 8471, 8473) is not publicly documented with precision. The Customs Tariff 2024-25 exists but specific rates and available exemption precedents are not clearly accessible. Investors require explicit clarity on:

1. the current baseline import duty rate (stated in HS code terms)
2. documented examples of sector-specific exemptions granted in recent years (renewable energy, semiconductors, telecom infrastructure, etc.) and the approval process
3. administrative timeline and procedural requirements for requesting a sector-specific exemption.

A sector-specific customs exemption for compute hardware imported for export-oriented compute services is implementable under existing executive authority and can be approved within a single fiscal cycle. Nepal has precedent: IT services received a 0% tax holiday under the IT Corridor Directive; renewable energy equipment receives tariff preferences.

Institutional Gap 4: Cross-Border Data Flows and Governing Law are Unspecified

The Directive 2081 establishes domestic data handling requirements. However, it does not address a fundamental question for export-oriented infrastructure: which jurisdiction's law governs a contract between a foreign AI lab and a Nepali data center operator when data is subject to multiple legal regimes?

Consider a practical scenario: A US-based AI lab contracts with a Nepal facility to train a model using a dataset containing European-sourced personal data (GDPR-regulated). The data is transmitted to Nepal, processed, and output is transmitted back to the US.



If unauthorized access occurs in Nepal, which privacy law governs the breach investigation—the Nepal Privacy Act 2075, the EU GDPR, or both? Which institution investigates? Can the AI lab pursue claims in international arbitration, or must it use Nepal's domestic courts?

The Privacy Act 2075 establishes privacy rights but does not clarify or provide explicit operational guidance regarding the treatment of foreign-origin personal data processed within Nepal. The Directive 2081 is silent on cross-border data transfers, data localization, and contractual choice-of-law. A multinational operator will not place sensitive workloads in Nepal without contractual language specifying:

1. which jurisdiction's substantive law governs data handling; (b) the mechanism for international data transfers
2. recourse if data inadvertently leaves Nepal's borders
3. whether Nepal recognizes international arbitration (ICC, UNCITRAL) for dispute resolution.

Singapore and Ireland both addressed this explicitly in their data center frameworks. Bhutan's framework includes choice-of-law provisions for international contracts. Nepal has not.

Why These Gaps Matter: Three Governance Practices Require Operationalization

Nepal has a policy framework for data centers. What it lacks is operational clarity on the four domains where foreign investors will conduct due diligence and where the facility will operate after launch. The [Public Institutions in the Age of AI: Emerging Practice](#) identifies several practices that governments must operationalize. Out of the seven practices identified, three are directly relevant to Nepal's institutional readiness.

Practice 1: Establishing Transparent Oversight.

Nepal needs a named, empowered Data Protection Authority with published breach investigation procedures, customer notification protocols, and timeline commitments. This authority must be credible and visible. Citizens and foreign customers have high expectations when technology governs their data. The framework emphasizes: "In the absence of visible and credible oversight, technology in government runs the risks of being perceived as arbitrary, unaccountable, and potentially unjust."



Practice 3: Building the Data Landscape.

Nepal needs clarity on data governance for export-oriented compute: data residency requirements, standards for international data transfers, and frameworks for cross-border data flows. Currently, the Directive 2081 specifies that data must be stored in Nepal but is silent on how foreign-held data transiting through Nepal is governed, what standards apply to cross-border transfers, and how the facility's data infrastructure interoperates with Nepal's broader data governance framework. Without this clarity, the facility's operational relationship to Nepal's data landscape is ambiguous.

Practice 6: Investing in Long-Term Outcomes.

Nepal needs to build workforce and institutional capacity for sustained operations, not just launch. The Public Workforce Performance and Prosperity report notes that "governments are underinvesting in the foundational and advanced digital skills necessary to effectively deploy technologies in the public sector." For compute export, Nepal's Ministry of Communication and Information Technology will need dedicated personnel with advanced capabilities in digital procurement and contract management (negotiating complex international PPAs and data processing agreements); technology infrastructure oversight (understanding GPU compute architectures, cooling systems, power distribution); international standards compliance (GDPR, SOC 2, ISO 27001); and contract dispute resolution. Building this capacity within the public sector is achievable but requires dedicated investment in recruitment and training. Without it, the facility will face operational and contractual risks after launch.

The data center licensing framework (Directive 2081) was designed to govern domestic government use of data center services. Export-oriented compute infrastructure serving international AI labs has fundamentally different requirements: international contract law, multi-jurisdictional compliance, 24/7 uptime guarantees, transparent data protection oversight, and specialized workforce capacity.

Implication: All Three Must Align

These gaps do not carry equal weight for all stakeholders, but must all resolve for viability. A customs exemption without firm power is operationally unusable. Firm power without DPA clarity is unacceptable to foreign customers. DPA clarity without customs exemption leaves the facility financially unviable.

Global Outlook

Across industry and policy analyses, there is broad agreement that AI workloads will account for an increasing share of total data center load. However, there remains significant uncertainty regarding the magnitude and trajectory of this demand.

We compile some of these industry estimates below for global energy use and India energy use, to highlight the range of these disagreements:

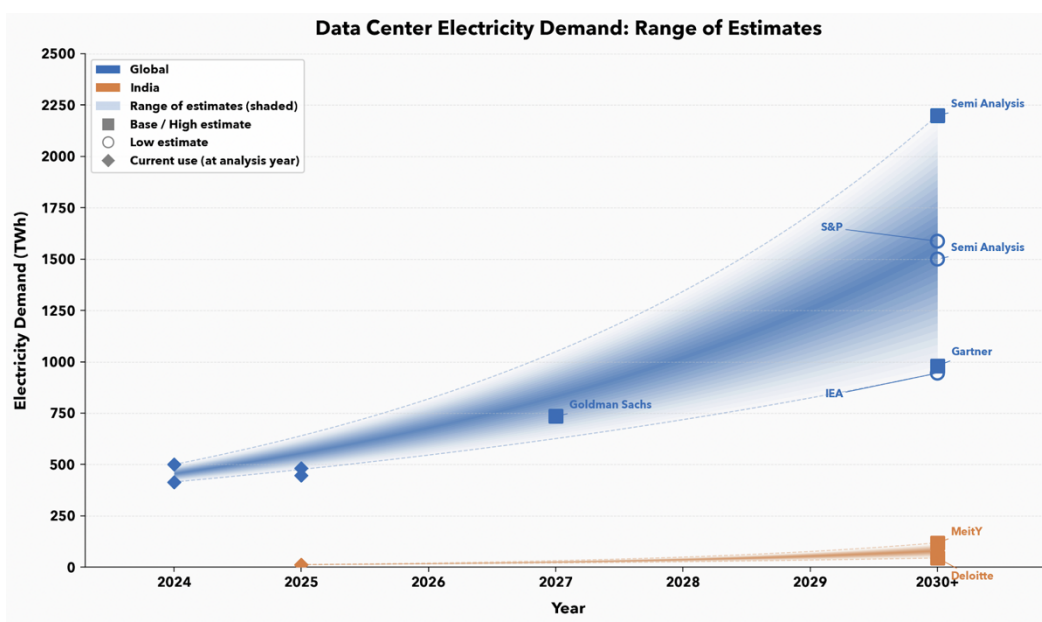


Figure 3: Estimates of AI Electricity Demand. Source: Author compilation from various reports.

Estimates vary widely due to differing assumptions about technological progress and adoption. The central tension lies between:

- **Efficiency gains:** Improvements in semiconductor design, model architectures, software optimization, power delivery systems, and cooling technologies
- **Demand expansion:** Growth in AI applications, longer context windows, increased use of reasoning-time compute, and the emergence of physical and world-model training.

On the supply side, both energy availability and advanced semiconductor capacity have emerged as binding constraints on AI development, particularly for large-scale model training. In response, major hyperscale data center investments are being planned and deployed globally, including in the United States and India. The [graph below from epoch.ai](https://epoch.ai) demonstrates the scale of the buildouts planned in the near-term.

● Data Centers

The largest known AI data center has computing power equivalent to 600,000 NVIDIA H100 chips.

Meta's Prometheus cluster has an estimated 700 MW of power capacity and \$20 billion in capital costs, making it the largest known AI data center. Microsoft's Fairwater Wisconsin data center is expected to be nearly 10x more powerful, at 5.2M H100-equivalents by September 2027.

[Learn more →](#)

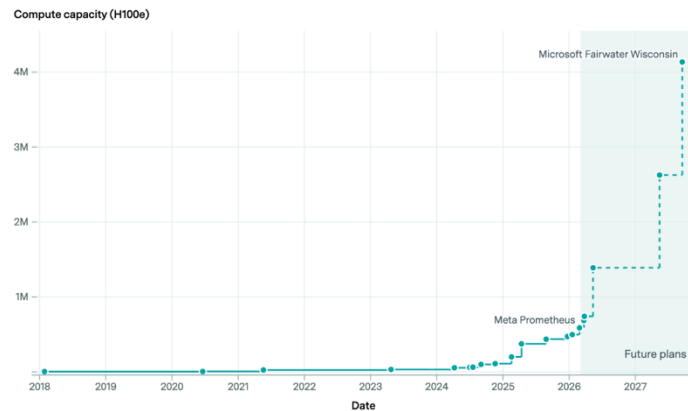


Figure 4: Epoch AI's Estimation of Compute Supply

These developments underscore both the capital intensity and the scale at which compute infrastructure is expanding, as well as the strategic importance countries are placing on securing energy and compute capacity.

This global context highlights a critical point for Nepal: the country cannot build first and solicit business later. It must work directly with AI Labs or global tech giants for direct, long-term offtake agreements that derisk demand volatility.



The Case for Nepal as a Compute Export Hub

Nepal presents a distinctive opportunity to capture a share of surging global compute demand, underpinned by three interlocking tailwinds.

1. **The Clean Energy Premium:** A structural shift has been underway in how hyperscale operators procure power. [Microsoft signed a 20-year power purchase agreement](#) with Constellation Energy to restart Three Mile Island's Unit 1 reactor to supply its data centers with carbon-free electricity. [Google partnered with Fervo Energy](#) to develop an enhanced geothermal power project, with carbon-free electricity now flowing to its Nevada data centers. Most recently, Google also struck a deal with NextEra to restart the Duane Arnold nuclear plant in Iowa. These transactions demonstrate that the world's largest compute buyers will pay a considerable premium for clean power. Nepal's run-of-river hydroelectricity fits precisely this profile.
2. **Cooling Efficiency:** Data center efficiency is measured by Power Usage Effectiveness (PUE), the ratio of total facility energy to energy delivered to IT equipment. In a hot or humid climate, mechanical chilling systems must work continuously, and even modern well-designed facilities [typically achieve PUEs in the 1.2–1.4 range](#). Nepal's naturally low ambient temperatures and access to glacial water, allow operators to substitute expensive mechanical cooling with outside air and water cooling for large portions of the year.
3. **Electricity Arbitrage:** Average [US residential electricity prices reached approximately 19 c / kWh by end of 2025](#), with a further increase projected in the coming years. Even on preferential industrial tariffs, large US data center operators pay between 5 and 17 cents per kWh depending on state, with a national commercial average in the 7–10 c range ([Source](#)). Nepal's regulated industrial tariff is considerably below this, enabling the country to export processed computation at global market rates.
4. **A Converging Policy Mandate:** This commercial logic is increasingly embedded in Nepal's political mainstream. The Digital Nepal Framework 2.0 identifies data centers and IT parks as national infrastructure priorities. More significantly, the Rastriya Swatantra Party has made the technology sector a central economic pillar. The party explicitly proposes promoting Nepal as a data center hub powered by



green energy, framing the goal as exporting "computation power" rather than raw electricity.

What to Build: Anticipating the Next Wave of Compute Demand

Despite the tailwinds discussed above, we must ensure that infrastructure built today can serve workloads that are still emerging. Based on our conversations with representatives from OpenAI, Google DeepMind, Waymo, Amazon, and xAI, three trends are reshaping compute demand beyond LLMs:

- 1) **World Models** are AI systems trained to predict physical dynamics from video and interaction data, learning how the world evolves given a state and an action, rather than predicting the next token in a text sequence.
- 2) **Physical AI** refers to AI systems that act in the real world. Reinforcement learning teaches these machines skills in simulated environments before real-world deployment, allowing them to learn safely through millions of trial-and-error tasks.
- 3) **Synthetic Data Generation** uses an already-trained model to produce training data for the next generation, rather than relying on human-generated content.

These three trends share a compute signature distinct from LLM workloads: massive video datasets, large-scale physics simulation, and closed-loop training where model outputs feed back into the next training round.

Table 4: Future AI Use Cases. Source: Authors

AI Compute Trends Headwinds & Tailwinds				
Technology / trend	Impact on Energy Demand	Power profile	Direction	Nepal Context
Training Workloads				
World model training	Training on video frames, 3D scenes, and sensor streams requires 10–100× more compute than equivalent text models. Hyperscalers in the US and India need new clean power locations urgently. Clean energy sourcing commands a price premium.	Sustained 50–500 MW loads for weeks at a time. Ideal for baseload hydro: no intermittency. GPU clusters run at ~95% utilisation during a training run.	Strong tailwind	Low-latency fiber to Mumbai/Singapore (~100 Gbps). Also requires 100+ MW stable supply per campus; needs Upper Tamakoshi-scale projects online.
Physical AI	Training physical-world models requires running millions of simulated robot-hours in parallel: pure GPU compute, no human in the loop. Simulation is largely parallel: suits distributed DC build-out rather than one giant cluster.	Similar energy profile to world models. Parallel structure means Nepal can serve this with multiple mid-size DC campuses.	Tailwind	Talent to operate facilities. GPU import duty regime must be competitive with Singapore and India.
Synthetic data generation	Producing synthetic training datasets means running large teacher models 24/7 to generate data for smaller student models. Completely location-agnostic: needs cheap clean electricity and bandwidth.	Sustained medium-to-high load, 24/7, highly predictable. Excellent match for run-of-river hydro: better load factor than bursty training jobs.	Strong tailwind	Bandwidth and reliability more critical than latency. Lower barrier to entry than frontier training: realistic near-term export opportunity.
Efficiency Trends				
Small language models + edge	If capable 3–7B models run on smartphones, a fraction of inference never touches a data center. Reduces aggregate inference DC demand growth.	Milliwatt-scale on device. Zero DC load for these workloads. Reduces inference DC demand only.	Headwind (inference only: training thesis unaffected)	Jevons paradox: Every prior efficiency wave in compute produced net more total energy consumption, not less. If Nepal builds training campuses, hardware refresh cycles must be planned carefully. Stranded asset risk if depreciation assumptions are wrong.
Quantisation + specialised silicon (3–10× efficiency per FLOP)	More efficient hardware reduces energy per operation. Slows growth in total DC energy demand at the margin. But every prior efficiency wave in computing has been absorbed by demand growth.	Reduces kWh per operation. Likely offset by demand expansion (Jevons). Net effect on total DC energy growth: probably neutral given historical pattern.	Mixed (per-unit efficiency up; total demand still grows)	

Notes: IEA DC electricity projection from 'Electricity 2025' (<https://www.iea.org/reports/electricity-2025>). Scaling law trajectory from Kaplan et al. (2020) and Hoffmann et al. 'Chinchilla' (2022). Synthetic data framing from Anthropic, OpenAI, DeepMind public technical reports 2023–2024. Jevons paradox argument: Sorrell (2009) 'Jevons' Paradox revisited', Energy Policy 37(4):1456–1469.



Nepal's opportunity is primarily in training, with potential for inference supply. Providing inference compute globally is constrained by latency requirements, grid reliability, data sovereignty laws, and sovereign risk. Meanwhile training jobs are latency-tolerant, geographically flexible, and run continuously for weeks or months. They are exactly the workloads that benefit most from cheap, clean, baseload power. Nepal does not need to compete for real-time serving next to end-users in Mumbai or Shanghai. It must play its advantage for the long-horizon training jobs that labs will need to run at scale over the next decade.



Financial Modeling

Below, we describe a simple financial model for a proposed GPU compute facility in Nepal, structured as a 512-GPU NVIDIA H100 cluster (64 DGX nodes). The expected load from this setup would approach 0.6 MW. We model three policy variables adding up to 18 scenarios (discussed below) as well as four electricity tariff levels — NRS 5/kWh (low case), NRS 7/kWh (medium), NRS 7.86/kWh (current industrial tariff, base case), and NRS 10.16/kWh (commercial tariff) — yielding 72 scenarios in total.

Total capital expenditure is estimated at \$41.9 million, comprising GPU server hardware (\$25.6m), data center infrastructure and installation/land (\$7.2m), and import duties (\$9.1m at a 35.6% effective rate). The model runs over a four-year useful life — matching the assumed economic lifespan of the H100 GPU — and applies a terminal exit at a 6.5× EBITDA multiple in Year 4.

Under the base case (Scenario 18: 25% corporate tax, 20% customs, 8% cost of debt, NRS 7.86/kWh industrial electricity tariff), the project returns a negative NPV of −\$2.5 million and an IRR of 9.2%, well below the 11.8% WACC. The model demonstrates that financial feasibility is primarily driven by import duty, not electricity cost, concessional financing or tax holidays — a finding that holds across all 72 scenarios.

Capital Expenditure Assumptions

We price each 8-GPU DGX H100 node at \$400,000, the midpoint of the \$373K–\$460K range observed across multiple 2025 vendor and reseller sources for a complete NVIDIA DGX H100 system (8× H100 SXM5, NVSwitch fabric, dual Xeon CPUs, NVMe storage, NVIDIA enterprise support, InfiniBand networking). The 64-node cluster totals \$25.6 million in raw equipment cost. Infrastructure costs are derived from the [Turner & Townsend 2025 Data Centre Cost Index](#), using an air-cooled base of \$10.5m / MW (benchmarked to Lagos as a comparable emerging-market location) with a 10% liquid-cooling premium, yielding \$11.55m/MW.

Import costs are the largest single risk factor. The model applies a 35.6% effective import duty rate, which combines a base customs duty (estimated between 5–20% under HS 8471/8473 for data processing machines) with a 13% VAT. This adds \$9.1 million to the capital stack — more than the entire \$7.2m infrastructure build cost, and roughly 22% of total project CAPEX.

Table 5: Capital Expenditure Assumptions. Source: Authors

Component	Amount (\$)
DGX H100 SuperPOD (64 nodes, 512 GPUs)	\$25,600,000
Infrastructure & Installation (DC shell, cooling, land)	\$7,194,715
Import Duties & Taxes (35.6% effective rate)	\$9,113,600
Total CAPEX	\$41,908,315

Revenue Assumptions

Revenue is modelled as hourly GPU rental income. The initial rate of \$3.50 per H100-hour is set above the spot market (\$2.50/hr on platforms like vast.ai) to reflect longer-term take-or-pay contracts with AI labs. Price declines at 10% per annum to capture the competitive effect of newer GPU generations entering the market over the projection horizon.

Utilization starts at 80% in Year 1 and declines by 10 percentage points annually, reaching 50% in Year 4. This trajectory is deliberately conservative, reflecting the real risk of hardware obsolescence as next-generation GPUs become commercially available during the model horizon. The combined effect of declining price and utilization causes revenue to fall from \$12.6 million in Year 1 to \$5.7 million in Year 4: a 54% decline.

Operating Costs

Annual maintenance and personnel costs are benchmarked at \$2,500 per GPU per year (from [GMI Cloud's 2025 data centre cost survey](#)), starting at \$1.28 million and growing at 5% annually for inflation. In the base case, energy costs use the Nepal Electricity Authority's time-of-day tariff for high-voltage industrial consumers, weighted across the three TOD windows that a 24/7 facility necessarily incurs: peak (6h at NPR 10.1/kWh), normal (12h at NPR 8.3/kWh), and off-peak (6h at NPR 4.75/kWh). The 24/7-weighted average is NPR 7.86/kWh, or roughly \$0.05/kWh at NPR 152/\$. Similarly, we consider the commercial tariffs and take a weighted average across three TOD windows. This is among the lowest grid electricity rates available to AI infrastructure globally.



Cost of Capital

The cost of equity is derived via a full CAPM build-up, from the perspective of a foreign investor:

- 4.4% USD risk-free rate (10-year Treasury, March 2026)
- 8.89% Nepal equity risk premium ([Damodaran, January 2026](#))
- levered beta of 2.34 (derived from an [unlevered beta](#) of 0.85 for a REIT/computer-peripherals blend, relevered at 70% debt and the applicable tax rate).

This yields a cost of equity of 25.2%, which is high but consistent with frontier-market greenfield infrastructure.

The cost of debt is scenario-dependent:

- 8.0% in the base case (matching Nepal's weighted average commercial lending rate per [NRB data](#))
- 6.5% in the concessional case (benchmarked to the IFC's [recent broadband infrastructure financing in Nepal](#) and green bond issuances supported by GGGI)

The resulting WACC ranges from approximately 10.5% to 11.8% depending on the scenario.

Depreciation, Debt, and Exit

The 512 GPUs are depreciated straight-line over four years (25% per annum), matching the assumed economic life. For simplicity, we do not model a second-generation refresh (e.g. purchasing Blackwell GPUs in Year 5).

Debt is structured as a four-year fully amortizing loan at 70% of total CAPEX (\$29.3 million), with equal annual payments. The exit in Year 4 applies a $6.5\times$ EBITDA multiple to terminal EBITDA, derived from a 50% discount to the [GDS Holdings ABS transaction](#) ($13.3\times$ EV/EBITDA, March 2025) to account for scale, illiquidity, obsolescence risk, and frontier-market discount, partially offset by a ESG premium for 100% hydropower operations.

Projected Cash Flows (Base Case)

The table below shows the revenue and cash-flow trajectory under the base-case assumptions before scenario adjustments are applied. Note that the Year 4 FCFF includes the terminal exit value (Year 4 EBITDA $\times$ $6.5\times$).

Table 6: Projected Cash Flows. Source: Authors

Year	Rate (\$/hr)	Utilization	Revenue (\$m)	EBITDA (\$m)	FCFF (\$m)
1	\$3.50	80%	\$12.6	\$11.0	\$11.0
2	\$3.15	70%	\$9.9	\$8.3	\$8.3
3	\$2.84	60%	\$7.6	\$6.0	\$6.0
4	\$2.55	50%	\$5.7	\$4.0	\$30.1

EBITDA margins are initially very strong due to the low variable cost structure: cheap energy is the facility's primary structural advantage. However, margins compress sharply as revenue declines while maintenance costs continue to grow. The bulk of the project's value recovery depends on the terminal exit, which alone represents more than half of total undiscounted cash inflows. This makes exit-multiple assumptions unusually consequential – a topic we return to in the sensitivity discussion.

Scenario Analysis

The model tests 72 scenarios across a $3 \times 3 \times 2 \times 4$ matrix: three corporate tax rates (0%, 20%, 25%), three customs duty rates (0%, 5%, 20%), two costs of debt (6.5%, 8.0%), and four electricity tariff levels. The table below presents results for the base-case tariff of NRS 7.86/kWh (current industrial rate) to maintain comparability with prior analysis; cross-tariff results are presented in the key findings and figures below.

- The corporate tax tiers correspond to a full tax holiday (0%), the IT-industry concession rate (20%), and the standard rate (25%)
- The customs tiers represent ITA accession or duty waiver (0%), low duty (5%), and the current effective rate (20%)
- The two cost-of-debt levels reflect concessional DFI/green-bond financing (6.5%) and standard commercial bank lending (8.0%).
- The four electricity tariff levels correspond to NRS 5/kWh (a potential low-tariff incentive), NRS 7/kWh (medium case), NRS 7.86/kWh (weighted average of current industrial 66 kV TOD rate, base case), and NRS 10.16/kWh (weighted average of commercial 33 kV TOD rate currently offered to data center operators). Tariff results are presented in the key findings below.

The full results are presented below:

Table 7: Scenario Analysis (Base Tariff Case). Source: Authors

Scenario	Corporate Tax Rate	Customs	Cost of Debt	NPV (\$m) NRS 7.86 base	IRR
1	0%	0%	6.5%	\$2.1m	16.0%
2	0%	5%	6.5%	\$0.8m	14.4%
3	0%	20%	6.5%	-\$3.0m	10.0%
4	20%	0%	6.5%	\$3.2m	15.1%
5	20%	5%	6.5%	\$1.9m	13.6%
6	20%	20%	6.5%	-\$2.0m	9.3%
7	25%	0%	6.5%	\$3.5m	14.9%
8	25%	5%	6.5%	\$2.2m	13.4%
9	25%	20%	6.5%	-\$1.7m	9.2%
10	0%	0%	8%	\$1.2m	16.0%
11	0%	5%	8%	-\$0.1m	14.4%
12	0%	20%	8%	-\$3.9m	10.0%
13	20%	0%	8%	\$2.4m	15.1%
14	20%	5%	8%	\$1.1m	13.6%
15	20%	20%	8%	-\$2.8m	9.3%
16	25%	0%	8%	\$2.7m	14.9%
17	25%	5%	8%	\$1.4m	13.4%
18	25%	20%	8%	-\$2.5m	9.2%

Key Findings from the Scenario Matrix

Customs duty is the single most decisive lever, and this finding is robust across all 72 scenarios. Every configuration with 20% customs duty produces a negative NPV regardless of corporate tax rate, cost of debt, or electricity tariff. Conversely, a customs waiver (0%) produces a positive NPV in every scenario across all four tariff levels. The average NPV gap between the 0%-customs and 20%-customs tiers is approximately \$5

million, dwarfing every other variable. No combination of tariff discounts, tax holidays, or cheap financing can bridge this gap. Figure 5 illustrates this dominance across the customs dimension.

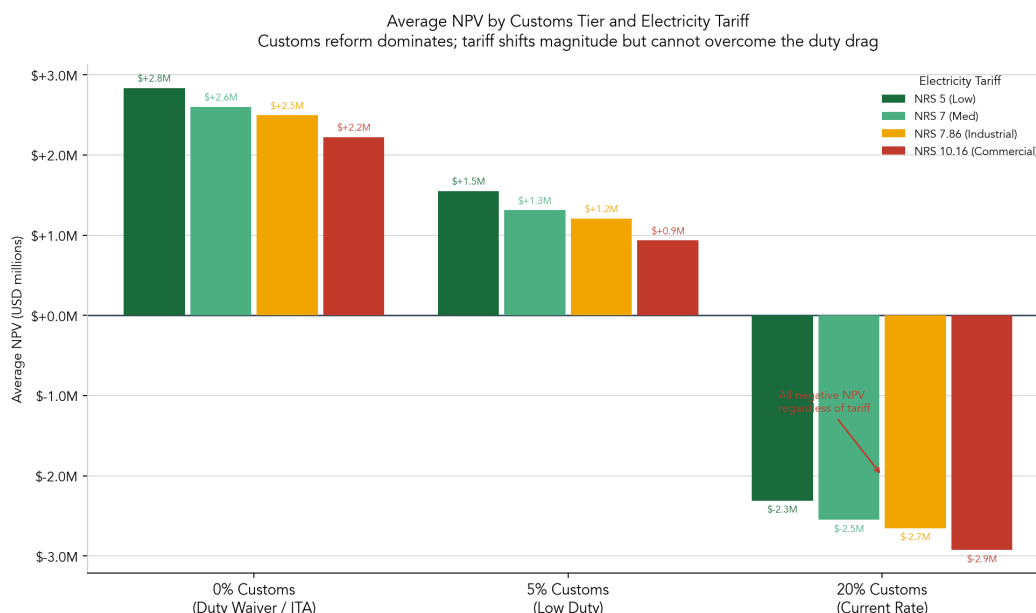


Figure 5: Average NPV by Customs Tier and Electricity Tariff. Source: Authors

A tax holiday is the wrong instrument and is in fact counterproductive. Across all 72 scenarios, the 0% corporate tax rate produces the lowest NPV of the three tax options, not the highest. The reason is the debt tax shield: with 70% leverage on \$29.3 million of financing, eliminating the tax deductibility of interest raises the effective cost of capital. At 0% tax with 8% debt, the WACC approaches 13.2% — well above the 11.8% at 25% tax — because none of the interest expense reduces tax liability. This higher discount rate outweighs the cash-flow benefit of paying no corporate tax, depressing NPV by \$0.5–\$0.9 million relative to the standard 25% rate across every scenario. Policymakers should not lead with tax holidays as the primary investment incentive: they are not only insufficient but actively harmful to project economics.

Electricity tariff has a meaningful but secondary role and acts as a decisive tipping variable in borderline configurations. Specifically, at 5% customs duty with 8% commercial debt financing, the project is marginally negative under both the industrial tariff (NRS 7.86, NPV ≈ −\$0.1m) and the commercial tariff (NRS 10.16, NPV ≈ −\$0.4m), but turns positive once the tariff falls to NRS 7 (NPV ≈ +\$0.03m) or NRS 5 (NPV ≈ +\$0.3m). This is the only configuration in which the tariff alone determines the viability of the project: all 20%-customs scenarios remain negative, and all 0%-customs scenarios

remain positive regardless of tariff. The commercial tariff of NRS 10.16/kWh should not be applied to export-oriented AI compute facilities.

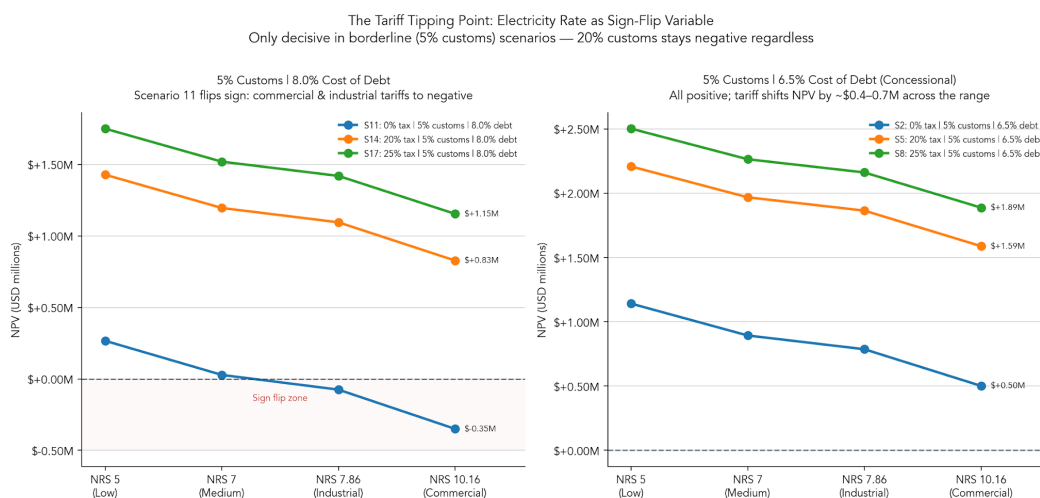


Figure 6: Tariff Tipping Point Analysis. Source: Authors

Concessional financing matters at the margin. The improvement from 8.0% to 6.5% cost of debt adds \$0.7–\$0.9 million in NPV across comparable scenarios — meaningful, but insufficient on its own to overcome a 20% customs rate. The combined effect of the three controllable levers — customs reform (20% to 0%), tariff optimization (NRS 10.16 to NRS 5), and concessional financing (8.0% to 6.5%) — lifts the project from a worst-case NPV of –\$2.7 million to a best-case of +\$3.8 million: a \$6.5 million swing in which customs reform accounts for approximately 79%, the electricity tariff for 9%, and concessional debt for the remaining 12%. Figure 7 decomposes this swing.

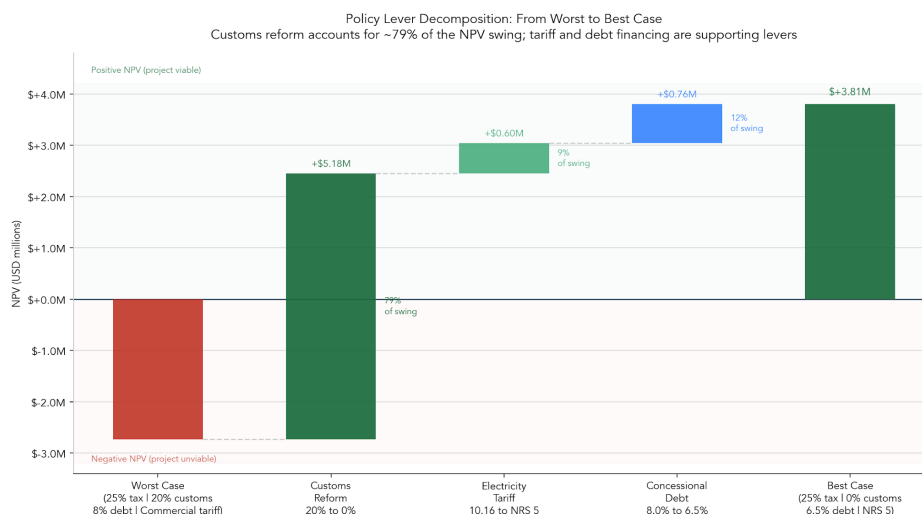


Figure 7: Policy Lever Decomposition

Policy Takeaways

- 1) **Customs Reform Is the Non-Negotiable Priority:** No combination of tax holidays, discounted electricity tariffs, or concessional debt can offset an import duty on \$25.6 million of GPU hardware. A sector-specific customs exemption for compute hardware imported for export-oriented services is the prerequisite for any viable project. Without it, nothing else moves the needle enough to matter.
- 2) **Avoid Tax Holidays as a Primary Incentive:** The scenario results are unambiguous: a full corporate tax holiday (0% rate) is counterproductive relative to both the 20% IT-sector rate and the 25% standard rate. The debt tax shield at 70% leverage dominates and eliminating it raises the effective WACC by approximately 150–200 basis points — worth \$0.5–\$0.9 million in lost NPV per scenario. If concessions are offered, the IT-sector rate of 20% is preferable to a full holiday, but neither tax instrument approaches the impact of customs reform. Investment incentive design should be reoriented accordingly.
- 3) **Concessional Financing and Industrial Tariff Protection Are Complementary Levers:** The 150-basis-point reduction in debt cost from 8.0% to 6.5% — achievable through IFC participation, green bond structures, or other development finance instruments — adds \$0.7–\$0.9 million in NPV per scenario. Equally important is the electricity tariff: the industrial rate (NRS 7.86/kWh) or below should be guaranteed for export-oriented compute facilities via a dedicated PPA. The commercial tariff of NRS 10.16/kWh reduces NPV by approximately \$280,000 per scenario relative to the industrial rate and, in borderline 5%-customs configurations, tips project viability from positive to negative.
- 4) **The Hydropower Advantage Is a Policy Variable, Not a Given:** Nepal's industrial electricity tariff of NRS 7.86/kWh (~\$0.058/kWh) is among the lowest rates available globally for commercial compute and is a genuine structural advantage. However, this advantage is not automatic: the commercial tariff of NRS 10.16/kWh being actively offered to some data center operators erodes this edge and, in borderline scenarios, tips projects into unviability. The government should formalize an industrial-rate PPA pathway for export-oriented GPU compute facilities — ideally at NRS 7/kWh or below to remain competitive with emerging jurisdictions such as Bhutan and the Gulf — and explicitly exclude commercial-tariff classification for this sector.



Conclusion

Nepal's case for hosting AI training compute rests on three structural advantages: clean firm power, low cooling overhead, and an industrial electricity tariff that is among the lowest available globally. The financial model is unambiguous: a 512-GPU pilot facility is commercially viable under a reformed policy regime and unviable under the current one. Every scenario in the sensitivity matrix that retains the 35.6% effective import duty produces a negative NPV. There is no margin for interpretation.

This clarity provides an evident advantage. Binding constraints with single levers are easier to address than diffuse ones. A sector-specific customs exemption for GPU hardware imported for export-oriented compute services can be issued under existing executive authority within a single fiscal cycle. It requires no new legislation, WTO accession, or constitutional change. Tax rebates and concessional DFI financing are valuable complements that improve signaling to international capital and bridge the cost-of-equity gap, but they cannot rescue the project on their own and should not be substitutes for customs reform.

However, customs reform alone is insufficient. Three institutional reforms must occur in parallel:

1. Operationalization of the Data Protection Authority: named institution designation, published breach investigation procedures, and explicit customer notification timelines (6-9 months)
2. Power Purchase Agreement framework for firm-power contracting: seasonal availability pricing, NEA capacity guarantees, and pricing differentiation between firm and interruptible power (4-6 months).
3. Cross-border data governance clarity: supplementary directive specifying choice-of-law provisions, international data transfer mechanisms, and recognition of international arbitration (2-3 months)

These are not optional refinements. None occurs without deliberate policy action. All three are implementable within 12-18 months through executive guidance and inter-agency coordination, without legislative delay. This window matters. The H100 generation that anchors the financial model in this paper has a finite commercial life — three to five years — before successor architectures (Blackwell, Rubin) compress GPU rental margins and restructure the global compute market. Bhutan, the Gulf states, and several Southeast Asian jurisdictions are actively positioning for the same opportunity with deeper sovereign capital and faster regulatory cycles. Nepal's cost advantage is



durable, but its first-mover position is not. The policy package that makes the pilot project viable — customs exemption, data governance laws, a PPA framework that prices in seasonal risk — is implementable within a single legislative session. Whether it is implemented within that window will determine whether the next decade of Nepal's hydropower surplus is exported as electricity at wholesale rates or as compute at retail margins. The economic case for the latter is, on the evidence of this paper, clear. The political case is the one that remains to be made.